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A Template-Matching-based Algorithm for Optical Microscope Image Stitching

Vikesh Singh Bhadouria*[1], You-rim Park[1], and Joo Beom Eom[1]
[1]*Department of Biomedical Science, Dankook University School of Medicine, Cheonan, South Korea*
Email: vikeshsb@dankook.ac.kr, jbeom@dankook.ac.kr

Abstract: This paper proposes a template-based method to enhance image-stitching in whole slide microscopy, tackling stage misalignment and flat field issues, improving accuracy and efficiency despite mechanical instability. © 2024 The Author(s)

1. Introduction

In the field of image processing techniques, the combining of numerous microscope images by image stitching is extremely important. This approach allows for the development of high-resolution composite images, which is critical in many scientific and medical fields. However, this amalgamation process faces significant challenges due to a variety of elements, most notably the stage's intrinsic mechanical instability and changes in illumination. To achieve perfect alignment, current approaches rely heavily on complicated feature detection and matching algorithms [1], [2], [3], [4], [5], [6]. These feature-based methods place heavy computational demands on matching several features across images to find their precise location and offset, especially when dealing with position-based whole slide microscopy (WSM) images [1], [3], [4], [5], [6], [7]. The absence of an appropriate approach for quickly identifying the translational offset between consecutive images while retaining computing efficiency is largely responsible for the very limited research on template matching for addressing image stitching and misaligned stages [8], [9], [10].

We suggest a simple image-processing approach that counts the number of peaks in the template's histogram as a way to work around this problem. If the histogram of the template shows more peaks in one template than the other, it makes sure that the template has a strong contrast. Combining microscope images taken under circumstances of small-stage misalignment is now computationally efficient and resilient due to this methodology, which lessens reliance on matching the complete image. The effectiveness of the template-based methodology is assessed via empirical investigations and comparisons with established approaches.

2. Experimental Setup and Method

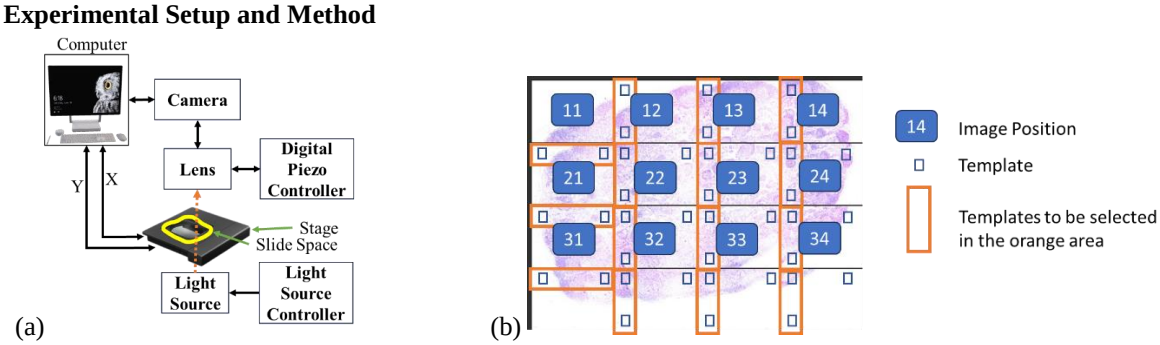

Fig. 1. (a) Schematic microscope image acquisition system [11], (b) A pictorial description of the template selection for various image positions

The purpose of the WSM imaging system, which is depicted in Fig. 1(a), is to enhance the efficiency of stitching even when minor stage misalignments are present. The system is comprised of four essential modules: a Piezo Stage, which is accountable for adjusting the objective lens's focus; a Motorized XY Stage, which facilitates the movement of the sample along two axes; Lighting, which utilizes a white LED to project a bright field light onto the sample; and a Camera Module, which is outfitted with a 20×/0.50 objective lens, a tube lens, and a camera with a 5.5 x 5.5 µm cell size. The acquisition procedure employs a piezo stage to adjust the focal plane and a motorized XY stage to facilitate slide movement along two dimensions, thereby guaranteeing in-focus images.

Template Selection and translational offsets: The template may be selected from a distance of several pixels from the image's boundary or at the boundary itself. We recommend that, in the proposed method, no more than three templates be selected for the "Image to be stitched" field. For matching with the neighboring image in the imaging

domain, the template whose histogram contains the greatest number of peaks will be selected. The evaluation will be conducted by comparing two scenarios according to the location of the "Image to be stitched," as depicted in Fig. 1(b).

After the template has been chosen, it is necessary to conduct template matching with the adjacent image. For example, in the case where a template is selected at position 12 on Image, it is necessary for the template chosen on this image to correspond with the image at position 11. In the same manner, image 21's template will correspond to image 11, and image 22's template will correspond to image 21, and so on. The offset of the subsequent image to be stitched onto the adjacent image can subsequently be calculated using this offset.

3. Results

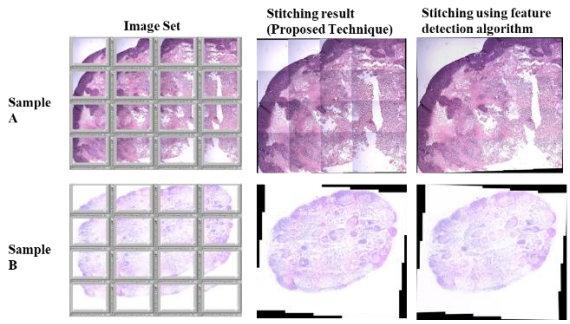

Fig. 2. Stitching results using the proposed method of template matching.

The results of template-matching-based image stitching method for position-based microscope imaging are presented here. The approach has been evaluated using microscope images and compared it to other image stitching methods. We measured computing efficiency by comparing our results to Microsoft ICE, a popular image-stitching method. Fig. 2 shows the method tested on two samples. Sample A shows minimal stage misalignment, whereas Sample B shows considerable stage misalignment. Fig. 2 shows the stitching results from our approach. Fig. 2 shows the translational offset of the image sets acquired using our recommended method compared to feature detection-based image stitching.

This approach stitches a single image in 8 seconds, considerably quicker than others. Recent research [12] shows that feature detection and matching require 89-95 milliseconds. Matching and stitching 16 images using such methods would take around 21 seconds. Due to our concise template for finding translational offsets between images, processing time is reduced.

4. Conclusion

In conclusion, the approach is designed for microscope images with translational offsets. Template matching in optical image processing reduces computational time for matching features with nearby images compared to feature detection algorithms. This method controls template size, affecting matching time. This work proves the accuracy of observed offsets between neighboring images for position-based image stitching in real time.

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